

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 8: Parameter Identifiability in the 1916 Model

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2026-10-19–2026-10-25

Overview

The historical eleven-parameter model is analyzed with a Fisher matrix to separate identifiable combinations from redundant notation.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$\text{rank}(F_{ij}) = \text{rank}\left(\sum_a \sigma_a^{-2} \partial_i T_a \partial_j T_a\right)$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$N_\theta = 11, N_{\text{bundle}} = 8$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$N_\theta^{2026} = 47, N_O^{2026} = 52$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper C, table 1
- SGT historical concordance, 1916–2026
- Hall, ch. 8